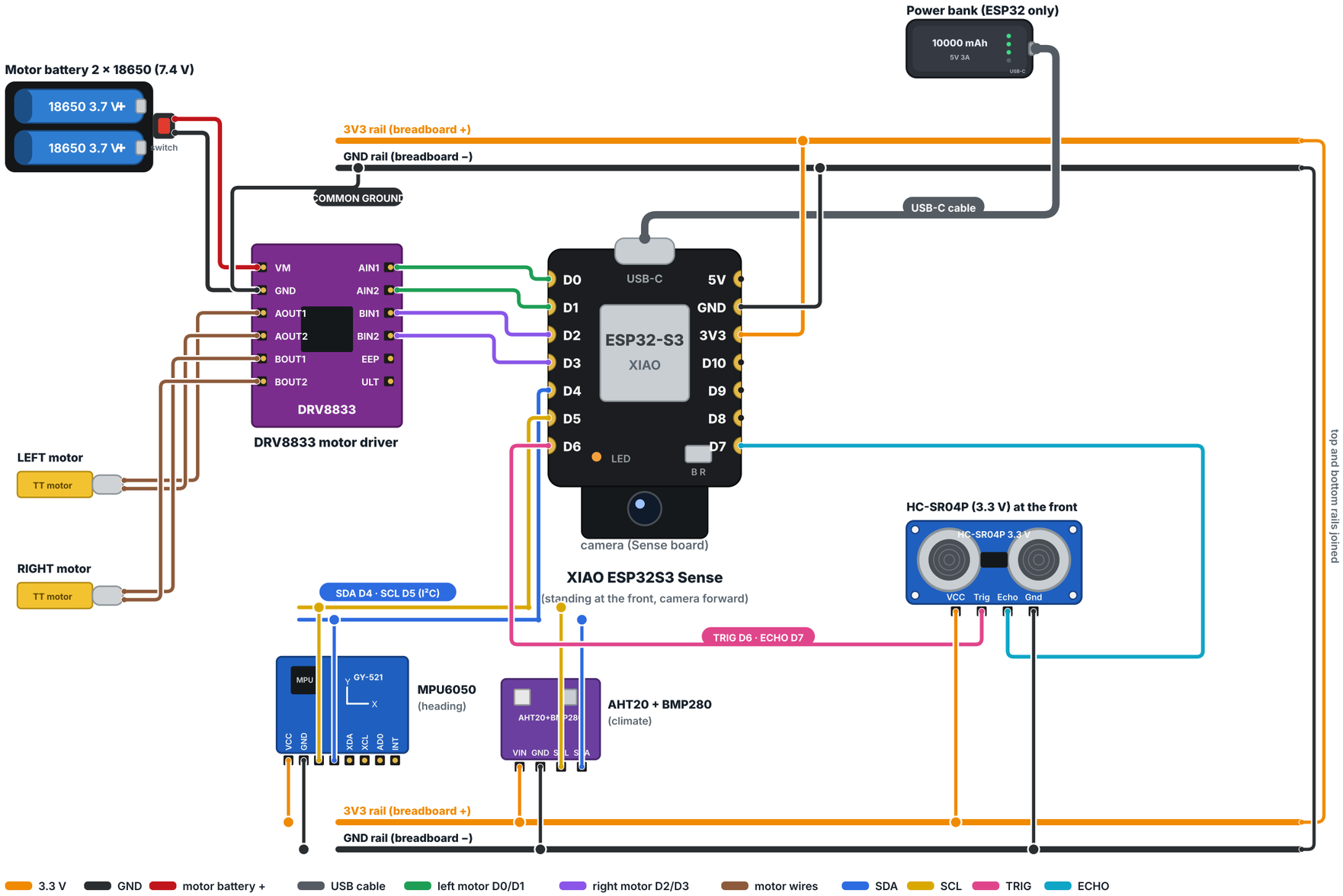
Raqib 3.0 · the whole robot's wiring

XIAO ESP32S3 Sense + DRV8833 + HC-SR04P + MPU6050 + AHT20/BMP280 on one half breadboard · about 22 jumper wires · no soldering except the XIAO's header pins



XIAO pins

Pin	Goes to	Why
D0	DRV8833 AIN1	left motor
D1	DRV8833 AIN2	left motor
D2	DRV8833 BIN1	right motor
D3	DRV8833 BIN2	right motor
D4	SDA → MPU6050 + AHT20/BMP280	I²C data
D5	SCL → MPU6050 + AHT20/BMP280	I²C clock
D6	HC-SR04P TRIG	distance
D7	HC-SR04P ECHO	distance
3V3	breadboard + rail	all sensors, DRV8833 EEP
GND	breadboard – rail	the one shared ground
USB-C	power bank	powers the XIAO only

Motor power

From	To	Note
2 × 18650 (7.4 V)	DRV8833 VM (+) and GND (–)	motors only
DRV8833 AOUT1/2	left TT motor	swap the two wires if it runs backwards
DRV8833 BOUT1/2	right TT motor	swap the two wires if it runs backwards
DRV8833 GND	breadboard – rail	joins the two batteries' grounds

3.3 V only on the XIAO. Never connect the motor battery or any 5 V signal to a XIAO pin. Use the HC-SR04**P** (3.3 V version).

Check it on the PC first: open the Wiring Bench (start_robot_sim.bat → http://localhost:8000/bench) and place the same wires; it tells you what would work before you plug anything in.